

P3-upq-Feb.-13KL-42 A4 D

Con. 6384-13.

GS-6105

(3 Hours)

[Total Marks : 100]

N.B. : (1) Question No. 1 is **compulsory**.(2) Attempt any **four** from remaining **six** questions.(3) Figures to the **right** indicate **full** marks.(4) Assume the **suitable** data if needed with **justification**.

1. (a) Find $L\{t e^{3t} \sin t\}$. 5
- (b) Show that every square matrix can be uniquely expressed as the sum of a Hermitian and skew- Hermitian matrix. 5
- (c) Find Z-transform and region of convergence of $f(k) = 3k, k \geq 0$. 5
- (d) Find the Fourier expansion of $f(x) = x^2$ where $-\pi \leq x \leq \pi$. 5
2. (a) Prove that following matrix is orthogonal and hence find its inverse. 6

$$A = \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}.$$

(b) Find $L^{-1} \left\{ \frac{s+2}{(s^2+4s+5)^2} \right\}$ 6

- (c) Obtain the Fourier expansion of $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in the interval and $0 \leq x \leq 2\pi$ 8
- and $f(x+2\pi) = f(x)$. Also deduce that,

(i) $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

(ii) $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots$

[TURN OVER

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3. (a) Investigate for what values of
- λ
- and
- μ
- the equations,
- 6**

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu \text{ have,}$$

- (i) No Solution
 (ii) a unique solution
 (iii) Infinite no. of solutions.

- (b) Obtain complex form of Fourier series for
- $f(x) = e^{ax}$
- $(-\pi, \pi)$
- where
- a
- is not an integer.
- 6**

- (c) Solve
- $(D^2 - D - 2)y = 20 \sin 2t$
- with
- $y(0) = 1, y'(0) = 2$
- .
- 8**

4. (a) Find Laplace transform of
- 6**

$$f(t) = a \sin pt \quad 0 < t \leq \pi/p$$

$$f(t) = 0 \quad \frac{\pi}{p} < t \leq \frac{2\pi}{p}$$

$$\text{and } f(t) = f\left(t + \frac{2\pi}{p}\right).$$

- (b) Find the inverse Z-transform for
- 6**

$$f(z) = \frac{1}{(z-3)(z-2)}$$

$$\text{for } 2 < |z| < 3.$$

- (c) Find inverse Laplace transform of
- 8**

$$(i) \quad \frac{e^{4-3s}}{(s+4)^{5/2}} \quad (ii) \quad \tan^{-1}\left(\frac{2}{s}\right)$$

5. (a) Examine whether the following vectors are linearly independent or dependent
- 6**
-
- $[2, 1, 1], [1, 3, 1], [1, 2, -1].$

- (b) Using Convolution theorem prove that
- 6**

$$\ell^{-1}\left[\frac{1}{s} \ell n\left(\frac{s+1}{s+2}\right)\right] = \int_0^t \left(\frac{e^{-2u} - e^{-u}}{u}\right) du$$

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(c) Using Fourier cosine Integral prove that

8

$$e^{-x} \cos x = \frac{1}{\pi} \int_0^{\infty} \frac{w^2 + 2}{w^4 + 4} \cos wx \, dw$$

6. (a) Find the Fourier Transform of $f(x) = e^{-1x}$ **6**(b) Find $z\{f(k)\}$ where $f(k) = \cos\left(\frac{k\pi}{u} + a\right)$ where $k \geq 0$.**6**(c) Find Fourier expansion of $f(x) = 2x - x^2$ where $0 \leq x \leq 3$ and period is 3.**8**

7. (a) Reduce the following matrix to normal form and find its rank.

6

$$A = \begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 5 & 3 & 3 & 11 \end{bmatrix}$$

(b) Evaluate $\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} \, dt$.**6**

(c) Show that the set of functions,

8

$$\sin\left(\frac{\pi x}{2L}\right), \sin\left(\frac{3\pi x}{2L}\right), \sin\left(\frac{5\pi x}{2L}\right), \dots$$

is orthogonal over $(0, L)$

Hence construct corresponding orthonormal set.
